

CHIRANJEEVI GUNDU

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SUMMARY

AI Engineer and systems architect with an M.S. in Computer Science (AI track) and 4 years taking AI from prototype to production. I design end-to-end GenAI systems — multi-agent orchestration, RAG pipelines, and model-routing layers — and ship them with backend rigor: typed APIs, sound data modeling, automated tests, CI/CD, and secure cloud and on-prem deployment on AWS and Azure. Equally comfortable translating ambiguous client requirements into scoped deliverables and debugging a retrieval miss three layers down. Proven record of measurable impact (40% faster triage, 60% faster queries, 10+ hours/week returned) and of building systems teams actually adopt and trust in production.

TECHNICAL SKILLS

Languages & ML: Python, SQL, JavaScript, TypeScript, Bash, PowerShell, PyTorch, Scikit-learn, pandas, NumPy, Text Classification

AI/LLM: OpenAI API, Anthropic Claude, LangChain, LangGraph, CrewAI, LlamaIndex, Hugging Face, RAG, Multi-Agent Orchestration, Tool-Calling, Prompt Engineering, Context Optimization, Fine-Tuning (LoRA, QLoRA), Embeddings, Semantic Search, Chunking & Retrieval Tuning, LLM Evaluation & Eval Sets, Guardrails AI, LiteLLM, Ollama, Llama 3, Mistral, OpenAI Vision, Agentic IDEs (Claude Code, GitHub Copilot, Cursor)

Backend & Data: FastAPI, Pydantic, REST APIs, Node.js, React, React Native, Microservices, Service-Oriented Architecture, PostgreSQL 16, pgvector, Pinecone, Redis, Row-Level Security, JWT Auth, Fernet Encryption, SQL Profiling (EXPLAIN/ANALYZE, pg_stat_statements), Indexing & Query Optimization, ETL & Ingestion Pipelines, IMAP, Microsoft Graph

Cloud & DevOps: AWS (EC2, S3, Lambda, ECS), Azure, Docker, Docker Compose, GitHub Actions, CI/CD, Git, PyTest, Plain-SQL Migrations (yoyo), Monitoring & Logging, Audit Trails, Tailscale, Agile/Scrum, Technical Documentation

PROFESSIONAL EXPERIENCE

AI Engineer

Aug 2025 - Present

Axitem Software Solution Inc. Remote, USA

- Deliver AI chatbots and document Q&A tools (Python, LangChain, OpenAI APIs) that answer client questions from internal documents in seconds instead of manual searching, now relied on daily by client support and operations teams.
- Ship RAG pipelines (Pinecone vector search, PostgreSQL) that ground every chatbot answer in real company documents with source citations, cutting hallucinations, improving reliability, and earning formal client project sign-off across engagements.
- Delivered RAG document-search chatbots to multiple enterprise clients, turning hours of manual document lookup into instant, cited AI answers embedded directly inside the existing portals their teams already used for daily work every single day.
- Automated multi-step business tasks such as data lookup and report drafting with LangChain and LangGraph agent workflows that plan steps, call tools, and verify their own output, removing hours of repetitive manual work for client teams.
- Unlocked privacy-sensitive enterprise deals by running open-source models (Llama 3, Mistral) fully in-house on client-owned infrastructure, keeping all client data off external APIs and satisfying strict internal security review requirements.
- Cut hallucinations with a three-layer trust design — answers grounded strictly in retrieved chunks, every claim linked to its source passage, and an explicit refusal path when retrieval confidence is low — so the assistant says "not in the documents" instead of guessing.
- Standardized delivery on a Dockerized FastAPI backend template with health endpoints and env-var configuration, making a move between AWS and Azure a configuration change rather than a rewrite and letting new client environments stand up in days.
- Turned every reported wrong answer into a permanent regression case by building evaluation sets from real user questions, then tuning chunking, embeddings, and retrieval against measured hit-rate rather than intuition.
- Clear client security review as an early gate by deploying inside the client's own cloud account and walking their IT team through every data flow before writing significant code — keeping projects on schedule instead of blocked late.
- Scoped an open-ended "ChatGPT but for us" request into a delivered phase 1 by running a working session on the top 20 real employee questions and demoing a prototype, converting a vague sponsor into the project's internal champion.
- Raised retrieval accuracy roughly 35% by scoring a fixed evaluation set before and after each change and separating retrieval failures from generation failures — the dominant fix was structure-aware parsing that stopped chunking from splitting tables mid-row.
- Keep indexes fresh with scheduled incremental re-ingestion that detects changed documents and upserts only those, versioning chunks so superseded content is removed rather than leaving two conflicting copies of the same policy in retrieval.

Systems Automation Engineer

Aug 2024 – May 2025

Sodexo Saint Louis, MO (on-campus role during Master's)

- Saved 10+ hours of manual work every week by automating routine system health checks and daily status reporting with scheduled Python and PowerShell scripts running across the entire campus kiosk and point-of-sale hardware fleet.
- Reduced campus kiosk downtime 15% by analyzing peak-usage logs and shifting the team from reactive break-fix repairs to proactive, scheduled maintenance carried out ahead of the busiest predicted high-traffic windows on campus each day.
- Documented and handed off every automation script so the team kept running them after the role ended, turning a temporary student position into durable operational tooling for campus digital technology.

Backend Software Engineer

Aug 2021 – Jul 2023

Tata Consultancy Services (TCS) Chennai, India

- Cut manual ticket-sorting time 40% for a Fortune 500 client by integrating an ML text-classification model into their IT support system through Python (FastAPI) REST APIs, replacing slow manual triage across busy, high-volume support queues.
- Improved mission-critical client workflow performance 60% by profiling and optimizing PostgreSQL queries, adding targeted indexes, and rewriting the inefficient backend data pipelines behind the client's most heavily used reporting workflows.
- Cut deployment failures 30% and held code coverage above 85% by building out thorough PyTest unit-test suites and enforcing CI/CD best practices on every single release shipped to enterprise clients across the delivery account each sprint.
- Made the ML routing system safe to switch on by designing a confidence-threshold fallback — low-confidence predictions returned to the manual queue rather than mis-routing — and logging every prediction against the human decision to measure live accuracy.
- Diagnosed performance bottlenecks with EXPLAIN/ANALYZE and pg_stat_statements rather than guesswork, then rewrote N+1 access patterns into joins and batch fetches and validated against production-scale data volumes before rollout.
- Translated business-language requirements from Fortune 500 retail and banking stakeholders into scoped technical proposals a non-engineer could approve, establishing the client-facing habit that now anchors my AI delivery work.

PROJECTS

Aadyon Assist — Open-Source AI Life-Ops Platform (github.com/chiranjeevigundu/aadyon-assist)

- Architected and open-sourced (MIT) a self-hosted, multi-user AI platform end to end — data model, agent layer, ingestion, REST API, and iPhone app — running as seven Docker Compose microservices on FastAPI and Postgres 16 with pgvector.
- Designed a multi-agent organization in which a CEO agent decomposes a goal and delegates to four specialist agent teams, bounding every run with explicit step limits and recording each step in an audit trail for full traceability and debugging.
- Engineered a human-in-the-loop safety model where read-only analysis runs autonomously while any action with a real-world side effect (money, email) becomes a proposal the user must approve, so no irreversible action ever executes on its own.
- Built a tiered model-routing layer over LiteLLM (OpenRouter, GPT-4o-mini, and local Llama 3 via Ollama) driven entirely by a database table, so cost, capability, and privacy each map to the right model with zero application code changes.
- Designed multi-tenant isolation using Postgres Row-Level Security enforced at the database, with JWT authentication and Fernet-encrypted credentials, so a single bug or injection can never leak one user's data to another user in the system.
- Built ingestion pipelines that turn emails, PDFs, and bank transactions into reviewable to-dos via LLM extraction and OpenAI Vision and shipped the whole system solo with 149 automated tests, GitHub Actions CI, and plain-SQL migrations.
- Modeled a "Digital Me" schema rolling profile, application, and milestone data into four transparent 0-100 life-dimension scores that each return their sub-components, so every number the system reports is auditable rather than opaque.
- Built read-only sync connectors over IMAP and Microsoft Graph with per-account cursors for incremental sync, routing all extracted items through an approval queue so extraction errors annoy the user instead of corrupting real records.
- Made a solo project survivable months later by concentrating 149 PyTest cases where determinism lives — schema and CRUD, auth and Row-Level Security boundaries, scoring math, API contracts — and testing around LLM calls rather than asserting on generations.
- Deployed the whole stack self-hosted and reachable only over a Tailscale network, with a daily briefing pushed to iPhone via self-hosted ntfy, making privacy structural rather than a policy promise.
- Reduced the operational burden of self-hosting with Docker Compose orchestration, a just task runner, and an automated backup service, keeping the same schema portable to a multi-tenant SaaS deployment since Row-Level Security was chosen with that path in mind.

EDUCATION

Saint Louis University, St. Louis, MO — M.S. Computer Science (Artificial Intelligence Track)

Aug 2023 – May 2025

Jawaharlal Nehru Technological University — B.Tech, Electronics & Communication Engineering

Jul 2017 – Aug 2021